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Problem

There are n people in a room. Some pairs of people are friends, and friendship is mutual. Alice and Bob play the following game. Alice moves first. On each turn, Alice chooses one person still in the room and removes that person. Suppose the removed person had exactly d friends among the people still in the room just before being removed. Then Bob must add exactly d new friendships between pairs of remaining people who are not already friends. If Bob cannot add exactly d new friendships, Alice wins. If, at any moment, every pair of remaining people are friends, Bob wins. Determine all initial friendship configurations for which Bob can win, no matter how Alice plays.

Solution. Consider a graph G where each vertex represents a person and v_i and v_j are adjacent only if the ith person and the jth person are friends. Since friendships are mutual, the edges are undirected.

We claim that Bob wins if and only if the number of total edges in the beginning was a triangular number less than or equal to $\binom{n}{2}$.

Claim 1 — The number of total edges in the graph is invariant.

Proof. Say there are k edges. If Alice removes one person with d friends, d vertices are removed. So we have $k-d$ edges. Now Bob must add d new friendships. So we have k edges again. $\square$

Observe that if Bob has won with m vertices in the graph, since all the edges are present, we have $\binom{m}{2}$ edges which is a triangular number.

If there were $\binom{k}{2}$ vertices in the beginning, we will show that Bob will win.

Say there are $m > k$ vertices and it is Alice's turn and she removes a vertex v with d edges. Consider the $m-1$ complete graph formed by the rest of the vertices. Since there are $\binom{m-1}{2} \geq \binom{k}{2}$ vertices in the complete graph and only $\binom{k}{2} - d$ are present, Bob can add at least d vertices.

Since the number of vertices are decreasing by 1, eventually there will be a stage where they will reach $m = k+1$ after which, when they will perform the steps

> mentioned above again, the inequality will achieve its equality case and it will be exactly a complete graph.